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SCIENCES

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01 October 2026

Dear Editor-in-Chief Dr. Vaughn Cooper,

Enclosed is our manuscript titled “Preventing the emergence of a zoonosis following eradication of an endemic pathogen”, which we are excited to submit to be considered for publication in *Evolution, Medicine, and Public Health*.

This study uses branching process models to explore an important question of general interest: how will the eradication of an endemic human pathogen (such as smallpox) potentiate the emergence of related zoonoses (such as mpox) in the human population?

Our models suggest the following :

1. We find that, in general, the probability of evolutionary emergence increases much steeper with increases in human-human transmission than was suggested by our earlier paper (Antia et al 2003 Nature). This is because we now consider how the number of mutations required for sustained human-human transmission decreases as the reproductive number approaches unity.
2. Previous studies on disease emergence (e.g. Plowright et al 2017 Nature Reviews Micro and Elby et al 2003 Nature) highlight the importance of preventing spillovers for the emergence of new pathogens. We find that this is not the case for the emergence of a zoonosis after eradication of an endemic relative. We show that, as population immunity to the endemic pathogen is lost, the probability of evolutionary emergence of the zoonosis is very weakly dependent on both the spillover rate and changes in the spillover rate and much more sensitive to increases in transmission in the human population.
3. We show that preventing the evolutionary emergence of the zoonotic disease will require a much lower level of vaccination compared with what will be required subsequent to its evolutionary emergence.

Our manuscript is for a broad audience. It advances our fundamental understanding of concepts in disease ecology related to the emergence of highly infectious pathogens from zoonotic reservoirs. We exceed the figure (by 1) and word (by 1,0000) counts slightly to discuss the potential application (and limitations) of our theoretical insights to an active public health concern – the emergence of mpox following smallpox eradication. All code will be made available via GitHub.

Thank you for considering our manuscript for publication and we look forward to hearing from you soon.

Sincerely,

Cora Hirst, on behalf of all coauthors: Julia Deichmann, Ananya Saha, Ira Longini, Andreas Handel, Marc Lipsitch, Daniel Weissman, and Rustom Antia.

We suggest the following potential reviewers:

1. Prof. Charles Nunn at Duke University (clnunn@duke.edu)
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